

## Chapter 10

# One- and Two-Sample Tests of Hypotheses

**Definition 10.1:** A **statistical hypothesis** is an assertion or conjecture concerning one or more populations.

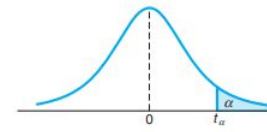
|                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach to Hypothesis Testing with Fixed Probability of Type I Error | <ol style="list-style-type: none"><li>1. State the null and alternative hypotheses.</li><li>2. Choose a fixed significance level <math>\alpha</math>.</li><li>3. Choose an appropriate test statistic and establish the critical region based on <math>\alpha</math>.</li><li>4. Reject <math>H_0</math> if the computed test statistic is in the critical region. Otherwise, do not reject.</li><li>5. Draw scientific or engineering conclusions.</li></ol> |
| Significance Testing ( $P$ -Value Approach)                           | <ol style="list-style-type: none"><li>1. State null and alternative hypotheses.</li><li>2. Choose an appropriate test statistic.</li><li>3. Compute the <math>P</math>-value based on the computed value of the test statistic.</li><li>4. Use judgment based on the <math>P</math>-value and knowledge of the scientific system.</li></ol>                                                                                                                   |

Table 10.3: Tests Concerning Means

| $H_0$                                   | Value of Test Statistic                                                                                                                                                                                                              | $H_1$                                                                      | Critical Region                                                                           |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| $\mu = \mu_0$                           | $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; \sigma \text{ known}$                                                                                                                                                                  | $\mu < \mu_0$<br>$\mu > \mu_0$<br>$\mu \neq \mu_0$                         | $z < -z_\alpha$<br>$z > z_\alpha$<br>$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$     |
| $\mu = \mu_0$                           | $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; v = n - 1,$<br>$\sigma \text{ unknown}$                                                                                                                                                     | $\mu < \mu_0$<br>$\mu > \mu_0$<br>$\mu \neq \mu_0$                         | $t < -t_\alpha$<br>$t > t_\alpha$<br>$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$                   | $z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}};$<br>$\sigma_1 \text{ and } \sigma_2 \text{ known}$                                                                                                | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $z < -z_\alpha$<br>$z > z_\alpha$<br>$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$                   | $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}};$<br>$v = n_1 + n_2 - 2,$<br>$\sigma_1 = \sigma_2 \text{ but unknown,}$<br>$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$                       | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $t < -t_\alpha$<br>$t > t_\alpha$<br>$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$     |
| $\mu_1 - \mu_2 = d_0$                   | $t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}};$<br>$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}};$<br>$\sigma_1 \neq \sigma_2 \text{ and unknown}$ | $\mu_1 - \mu_2 < d_0$<br>$\mu_1 - \mu_2 > d_0$<br>$\mu_1 - \mu_2 \neq d_0$ | $t' < -t_\alpha$<br>$t' > t_\alpha$<br>$t' < -t_{\alpha/2} \text{ or } t' > t_{\alpha/2}$ |
| $\mu_D = d_0$<br>paired<br>observations | $t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}};$<br>$v = n - 1$                                                                                                                                                                             | $\mu_D < d_0$<br>$\mu_D > d_0$<br>$\mu_D \neq d_0$                         | $t < -t_\alpha$<br>$t > t_\alpha$<br>$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$     |







**Table A.4** Critical Values of the  $t$ -Distribution

| $v$      | $\alpha$ |       |       |       |       |       |        |
|----------|----------|-------|-------|-------|-------|-------|--------|
|          | 0.40     | 0.30  | 0.20  | 0.15  | 0.10  | 0.05  | 0.025  |
| 1        | 0.325    | 0.727 | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 |
| 2        | 0.289    | 0.617 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  |
| 3        | 0.277    | 0.584 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  |
| 4        | 0.271    | 0.569 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  |
| 5        | 0.267    | 0.559 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  |
| 6        | 0.265    | 0.553 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  |
| 7        | 0.263    | 0.549 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  |
| 8        | 0.262    | 0.546 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  |
| 9        | 0.261    | 0.543 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  |
| 10       | 0.260    | 0.542 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  |
| 11       | 0.260    | 0.540 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  |
| 12       | 0.259    | 0.539 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  |
| 13       | 0.259    | 0.538 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  |
| 14       | 0.258    | 0.537 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  |
| 15       | 0.258    | 0.536 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  |
| 16       | 0.258    | 0.535 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  |
| 17       | 0.257    | 0.534 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  |
| 18       | 0.257    | 0.534 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  |
| 19       | 0.257    | 0.533 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  |
| 20       | 0.257    | 0.533 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  |
| 21       | 0.257    | 0.532 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  |
| 22       | 0.256    | 0.532 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  |
| 23       | 0.256    | 0.532 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  |
| 24       | 0.256    | 0.531 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  |
| 25       | 0.256    | 0.531 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  |
| 26       | 0.256    | 0.531 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  |
| 27       | 0.256    | 0.531 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  |
| 28       | 0.256    | 0.530 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  |
| 29       | 0.256    | 0.530 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  |
| 30       | 0.256    | 0.530 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  |
| 40       | 0.255    | 0.529 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  |
| 60       | 0.254    | 0.527 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000  |
| 120      | 0.254    | 0.526 | 0.845 | 1.041 | 1.289 | 1.658 | 1.980  |
| $\infty$ | 0.253    | 0.524 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  |



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**Example 10.3:** A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

- Solution:**
1.  $H_0: \mu = 70$  years.
  2.  $H_1: \mu > 70$  years.
  3.  $\alpha = 0.05$ .
  4. Critical region:  $z > 1.645$ , where  $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ .
  5. Computations:  $\bar{x} = 71.8$  years,  $\sigma = 8.9$  years, and hence  $z = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.02$ .
  6. Decision: Reject  $H_0$  and conclude that the mean life span today is greater than 70 years.

The  $P$ -value corresponding to  $z = 2.02$  is given by the area of the shaded region in Figure 10.10.

Using Table A.3, we have

$$P = P(Z > 2.02) = 0.0217.$$

As a result, the evidence in favor of  $H_1$  is even stronger than that suggested by a 0.05 level of significance. 

**Example 10.4:** A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that  $\mu = 8$  kilograms against the alternative that  $\mu \neq 8$  kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

- Solution:**
1.  $H_0: \mu = 8$  kilograms.
  2.  $H_1: \mu \neq 8$  kilograms.
  3.  $\alpha = 0.01$ .
  4. Critical region:  $z < -2.575$  and  $z > 2.575$ , where  $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ .
  5. Computations:  $\bar{x} = 7.8$  kilograms,  $n = 50$ , and hence  $z = \frac{7.8 - 8}{0.5/\sqrt{50}} = -2.83$ .
  6. Decision: Reject  $H_0$  and conclude that the average breaking strength is not equal to 8 but is, in fact, less than 8 kilograms.

Since the test in this example is two tailed, the desired  $P$ -value is twice the area of the shaded region in Figure 10.11 to the left of  $z = -2.83$ . Therefore, using Table A.3, we have

$$P = P(|Z| > 2.83) = 2P(Z < -2.83) = 0.0046,$$

which allows us to reject the null hypothesis that  $\mu = 8$  kilograms at a level of significance smaller than 0.01. 

**Example 10.5:** The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal.

- Solution:**
1.  $H_0: \mu = 46$  kilowatt hours.
  2.  $H_1: \mu < 46$  kilowatt hours.
  3.  $\alpha = 0.05$ .
  4. Critical region:  $t < -1.796$ , where  $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$  with 11 degrees of freedom.
  5. Computations:  $\bar{x} = 42$  kilowatt hours,  $s = 11.9$  kilowatt hours, and  $n = 12$ .  
Hence,

$$t = \frac{42 - 46}{11.9/\sqrt{12}} = -1.16, \quad P = P(T < -1.16) \approx 0.135.$$



**Example 10.6:** An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume the populations to be approximately normal with equal variances.

**Solution:** Let  $\mu_1$  and  $\mu_2$  represent the population means of the abrasive wear for material 1 and material 2, respectively.

1.  $H_0: \mu_1 - \mu_2 = 2$ .
2.  $H_1: \mu_1 - \mu_2 > 2$ .
3.  $\alpha = 0.05$ .
4. Critical region:  $t > 1.725$ , where  $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}$  with  $v = 20$  degrees of freedom.
5. Computations:

$$\begin{array}{lll} \bar{x}_1 = 85, & s_1 = 4, & n_1 = 12, \\ \bar{x}_2 = 81, & s_2 = 5, & n_2 = 10. \end{array}$$

Hence

$$\begin{aligned} s_p &= \sqrt{\frac{(11)(16) + (9)(25)}{12 + 10 - 2}} = 4.478, \\ t &= \frac{(85 - 81) - 2}{4.478 \sqrt{1/12 + 1/10}} = 1.04, \\ P &= P(T > 1.04) \approx 0.16. \quad (\text{See Table A.4.}) \end{aligned}$$

6. Decision: Do not reject  $H_0$ . We are unable to conclude that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units. 

**Case Study 10.1: Blood Sample Data:** In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the

first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given in Table 10.2.

Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes.

Table 10.2: Data for Case Study 10.1

| Deer | Androgen (ng/mL)     |                            | $d_i$ |
|------|----------------------|----------------------------|-------|
|      | At Time of Injection | 30 Minutes after Injection |       |
| 1    | 2.76                 | 7.02                       | 4.26  |
| 2    | 5.18                 | 3.10                       | -2.08 |
| 3    | 2.68                 | 5.44                       | 2.76  |
| 4    | 3.05                 | 3.99                       | 0.94  |
| 5    | 4.10                 | 5.21                       | 1.11  |
| 6    | 7.05                 | 10.26                      | 3.21  |
| 7    | 6.60                 | 13.91                      | 7.31  |
| 8    | 4.79                 | 18.53                      | 13.74 |
| 9    | 7.39                 | 7.91                       | 0.52  |
| 10   | 7.30                 | 4.85                       | -2.45 |
| 11   | 11.78                | 11.10                      | -0.68 |
| 12   | 3.90                 | 3.74                       | -0.16 |
| 13   | 26.00                | 94.03                      | 68.03 |
| 14   | 67.48                | 94.03                      | 26.55 |
| 15   | 17.04                | 41.70                      | 24.66 |

**Solution:** Let  $\mu_1$  and  $\mu_2$  be the average androgen concentration at the time of injection and 30 minutes later, respectively. We proceed as follows:

1.  $H_0$ :  $\mu_1 = \mu_2$  or  $\mu_D = \mu_1 - \mu_2 = 0$ .
2.  $H_1$ :  $\mu_1 \neq \mu_2$  or  $\mu_D = \mu_1 - \mu_2 \neq 0$ .
3.  $\alpha = 0.05$ .
4. Critical region:  $t < -2.145$  and  $t > 2.145$ , where  $t = \frac{\bar{d} - d_0}{s_D/\sqrt{n}}$  with  $v = 14$  degrees of freedom.
5. Computations: The sample mean and standard deviation for the  $d_i$  are

$$\bar{d} = 9.848 \quad \text{and} \quad s_d = 18.474.$$

Therefore,

$$t = \frac{9.848 - 0}{18.474/\sqrt{15}} = 2.06.$$

6. Though the  $t$ -statistic is not significant at the 0.05 level, from Table A.4,

$$P = P(|T| > 2.06) \approx 0.06.$$

As a result, there is some evidence that there is a difference in mean circulating levels of androgen. 



**Exercise questions (10.19 to 10.41)**

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**10.19** In a research report, Richard H. Weindruch of the UCLA Medical School claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that  $\mu < 40$  if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a  $P$ -value in your conclusion.

The hypotheses are

$$H_0 : \mu = 40 \text{ months,}$$

$$H_1 : \mu < 40 \text{ months.}$$

Now,  $z = \frac{38-40}{5.8/\sqrt{64}} = -2.76$ , and  $P\text{-value} = P(Z < -2.76) = 0.0029$ . Decision: reject  $H_0$ .

**10.30** A random sample of size  $n_1 = 25$ , taken from a normal population with a standard deviation  $\sigma_1 = 5.2$ , has a mean  $\bar{x}_1 = 81$ . A second random sample of size  $n_2 = 36$ , taken from a different normal population with a standard deviation  $\sigma_2 = 3.4$ , has a mean  $\bar{x}_2 = 76$ . Test the hypothesis that  $\mu_1 = \mu_2$  against the alternative,  $\mu_1 \neq \mu_2$ . Quote a  $P$ -value in your conclusion.

The hypotheses are

$$H_0 : \mu_1 = \mu_2,$$

$$H_1 : \mu_1 \neq \mu_2.$$

Since the variances are known, we obtain  $z = \frac{81-76}{\sqrt{5.2^2/25+3.4^2/36}} = 4.22$ . So,  $P\text{-value} \approx 0$  and we conclude that  $\mu_1 > \mu_2$ .

**10.23** Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the

contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

The hypotheses are

$$H_0 : \mu = 10,$$

$$H_1 : \mu \neq 10.$$

$\alpha = 0.01$  and  $df = 9$ .

Critical region:  $t < -3.25$  or  $t > 3.25$ .

Computation:  $t = \frac{10.06-10}{0.246/\sqrt{10}} = 0.77$ .

Decision: Fail to reject  $H_0$ .

**10.35** To find out whether a new serum will arrest leukemia, 9 mice, all with an advanced stage of the disease, are selected. Five mice receive the treatment and 4 do not. Survival times, in years, from the time the experiment commenced are as follows:

|              |     |     |     |     |     |
|--------------|-----|-----|-----|-----|-----|
| Treatment    | 2.1 | 5.3 | 1.4 | 4.6 | 0.9 |
| No Treatment | 1.9 | 0.5 | 2.8 | 3.1 |     |

At the 0.05 level of significance, can the serum be said to be effective? Assume the two populations to be normally distributed with equal variances.

The hypotheses are

$$H_0 : \mu_1 - \mu_2 = 0,$$

$$H_1 : \mu_1 - \mu_2 < 0.$$

$$\alpha = 0.05$$

Critical region:  $t < -1.895$  with 7 degrees of freedom.

$$\text{Computation: } s_p = \sqrt{\frac{(3)(1.363) + (4)(3.883)}{7}} = 1.674, \text{ and } t = \frac{2.075 - 2.860}{1.674\sqrt{1/4 + 1/5}} = -0.70.$$

Decision: Do not reject  $H_0$ .

**10.28** According to *Chemical Engineering*, an important property of fiber is its water absorbency. The average percent absorbency of 25 randomly selected pieces of cotton fiber was found to be 20 with a standard deviation of 1.5. A random sample of 25 pieces of acetate yielded an average percent of 12 with a standard deviation of 1.25. Is there strong evidence that the population mean percent absorbency is significantly higher for cotton fiber than for acetate? Assume that the percent absorbency is approximately normally distributed and that the population variances in percent absorbency for the two fibers are the same. Use a significance level of 0.05.

The hypotheses are

$$H_0 : \mu_C = \mu_A,$$

$$H_1 : \mu_C > \mu_A,$$

with  $s_p = \sqrt{\frac{(24)(1.5)^2 + (24)(1.25)^2}{48}} = 1.3807$ . We obtain  $t = \frac{20.0 - 12.0}{1.3807\sqrt{2/25}} = 20.48$ . Since  $P(T > 20.48) \approx 0$ , we conclude that the mean percent absorbency for the cotton fiber is significantly higher than the mean percent absorbency for acetate.



**10.32** *Amstat News* (December 2004) lists median salaries for associate professors of statistics at research institutions and at liberal arts and other institutions in the United States. Assume that a sample of 200 associate professors from research institutions has an average salary of \$70,750 per year with a standard deviation of \$6000. Assume also that a sample of 200 associate professors from other types of institutions has an average salary of \$65,200 with a standard deviation of \$5000. Test the hypothesis that the mean salary for associate professors in research institutions is \$2000 higher than for those in other institutions. Use a 0.01 level of significance.

The hypotheses are

$$\begin{aligned} H_0 : \mu_1 - \mu_2 &= \$2,000, \\ H_1 : \mu_1 - \mu_2 &> \$2,000. \end{aligned}$$

$$\alpha = 0.01.$$

Critical region:  $z > 2.33$ .

$$\text{Computation: } z = \frac{(70750 - 65200) - 2000}{\sqrt{(6000)^2/200 + (5000)^2/200}} = 6.43, \text{ with a } P\text{-value} = P(Z > 6.43) \approx$$

0. Reject  $H_0$  and conclude that the mean salary for associate professors in research institutions is \$2000 higher than for those in other institutions.

**10.42** Five samples of a ferrous-type substance were used to determine if there is a difference between a laboratory chemical analysis and an X-ray fluorescence analysis of the iron content. Each sample was split into two subsamples and the two types of analysis were applied. Following are the coded data showing the iron content analysis:

| Analysis | Sample |     |     |     |     |
|----------|--------|-----|-----|-----|-----|
|          | 1      | 2   | 3   | 4   | 5   |
| X-ray    | 2.0    | 2.0 | 2.3 | 2.1 | 2.4 |
| Chemical | 2.2    | 1.9 | 2.5 | 2.3 | 2.4 |

Assuming that the populations are normal, test at the 0.05 level of significance whether the two methods of analysis give, on the average, the same result.

The hypotheses are

$$\begin{aligned} H_0 : \mu_1 &= \mu_2, \\ H_1 : \mu_1 &\neq \mu_2. \end{aligned}$$

$$\alpha = 0.05.$$

Critical regions  $t < -2.776$  or  $t > 2.776$ , with 4 degrees of freedom.

$$\text{Computation: } \bar{d} = -0.1, s_d = 0.1414, \text{ so } t = \frac{-0.1}{0.1414/\sqrt{5}} = -1.58.$$

Decision: Do not reject  $H_0$  and conclude that the two methods are not significantly different.